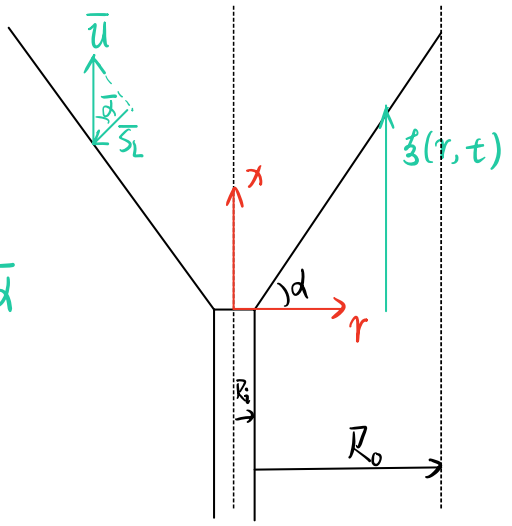
$$\frac{DG}{Dt} = 0$$

$$\Rightarrow \quad \frac{DG}{Dt} = \frac{\partial G}{\partial t} + (\vec{u} - S_L \vec{n}) \cdot \nabla G = 0$$

$$\Rightarrow \quad \frac{\partial G}{\partial t} + \vec{u} \cdot \nabla G = S_L |\nabla G|$$

V-flame :



$$\frac{\bar{S}_L}{\bar{u}} = \cos\bar{\alpha}$$

Let $G(x,r,t) = x - \xi(r,t)$

$$\Rightarrow \quad \vec{u} = (u(x,r,t),\ 0) \qquad \text{neglect } v$$

where $u(x,r,t) = u(\xi(r,t), r, t) = u(r,t)$

Similarly, $\phi = \phi(x,r,t) = \phi(r,t)$

Substitude into $\frac{\partial G}{\partial t} + \vec{u} \cdot \nabla G = S_L |\nabla G|$

$$\Rightarrow \quad -\frac{\partial \xi}{\partial t} + u = S_L \sqrt{1 + \left(\frac{\partial \xi}{\partial r}\right)^2}$$

$$\Rightarrow \quad \frac{\partial \xi}{\partial t} = u - S_L \sqrt{1 + \left(\frac{\partial \xi}{\partial r}\right)^2}$$

Decomposition :

$$\xi(r,t) = \bar{\xi}(r) + \xi'(r,t)$$

$$u(r,t) = \bar{u} + u'(r,t)$$

$$\phi(r,t) = \bar{\phi} + \phi'(r,t)$$

$$S_L(r,t) = \bar{S}_L(r) + S_L'(r,t)$$

$\bar{\xi}(r)$ → Can be calculated by steady G-equation

$\bar{u}$ → mean flow

$u'(r,t)$ → acoustic forcing (if exist)

$\phi'(r,t)$ → EQR forcing

(consider convective and diffusion)

global EQR (nominal)

Assumption :

$$u'(r,t) = \hat{u}(r)e^{i\omega t} = \varepsilon_u \bar{u}\, e^{-ik\bar{\xi}(r)} e^{i\omega t}$$

$$\phi'(r,t) = \hat{\phi}(r)e^{i\omega t} = \varepsilon_\phi \bar{\phi}\, e^{-(\sigma_c + ik)\bar{\xi}(r)} e^{i\omega t} = \varepsilon_\phi \bar{\phi}\, e^{-k_\phi \bar{\xi}(r)} e^{i\omega t}$$

where ε is perturbation level

$k \approx \frac{\omega}{\bar{u}}$ is the convective wavenumber (velocity from flame base and EQR)

σ_c represent the diffusion

$k_\phi = \sigma_c + ik$

① Steady equation : $\frac{\partial \bar{\xi}}{\partial t} = 0$

$$0 = \bar{u} - \bar{S}_L \sqrt{1 + \left(\frac{\partial \bar{\xi}}{\partial r}\right)^2}$$

$$\Rightarrow \quad \frac{\partial \bar{\xi}}{\partial r} = \sqrt{\left(\frac{\bar{u}}{\bar{S}_L}\right)^2 - 1} \qquad (\text{負值已舍})$$

$$\Rightarrow \quad \bar{\xi} = (r - R_i)\sqrt{\left(\frac{\bar{u}}{\bar{S}_L}\right)^2 - 1} \qquad \text{B.C. : } \bar{\xi}(R_i) = 0$$

$$= \frac{r - R_i}{\bar{S}_L}\sqrt{\bar{u}^2 - \bar{S}_L^2} = (r - R_i)\tan\alpha$$

② Linearized equation :

$$\frac{\partial \xi'}{\partial t} = u' - \bar{S}_L \frac{\partial \xi'}{\partial r}\frac{\partial \bar{\xi}}{\partial r}\left(1 + \left(\frac{\partial \bar{\xi}}{\partial r}\right)^2\right)^{-1/2} - S_L' \sqrt{1 + \left(\frac{\partial \bar{\xi}}{\partial r}\right)^2}$$

$$\sqrt{1+\left(\frac{\partial \bar{\xi}}{\partial r}\right)^2} = \frac{\bar{u}}{\bar{\zeta}_L}$$

$$\frac{\partial \bar{\xi}}{\partial r} = \sqrt{\left(\frac{\bar{u}}{\bar{\zeta}_L}\right)^2 - 1}$$

$$\frac{\partial \xi'}{\partial t} = u' - \bar{\zeta}_L \sqrt{\left(\frac{\bar{u}}{\bar{\zeta}_L}\right)^2 - 1}\ \frac{\bar{\zeta}_L}{\bar{u}}\ \frac{\partial \xi'}{\partial r} - \frac{\bar{u}}{\bar{\zeta}_L}\zeta_L'$$

$$\Rightarrow \quad \frac{\partial \xi'}{\partial t} - u' + \frac{\bar{u}}{\bar{\zeta}_L}\zeta_L' = -\frac{\bar{\zeta}_L}{\bar{u}}\sqrt{\bar{u}^2 - \bar{\zeta}_L^2}\ \frac{\partial \xi'}{\partial r}$$

Fourier transform in t $\longrightarrow$ $i\omega \hat{\xi} - \hat{u} + \frac{\bar{u}}{\bar{\zeta}_L}\hat{\zeta}_L = -\frac{\bar{\zeta}_L}{\bar{u}}\sqrt{\bar{u}^2 - \bar{\zeta}_L^2}\ \frac{\partial \hat{\xi}}{\partial r}$

$$\Rightarrow \quad \left(i\omega \hat{\xi} - \hat{u} + \frac{\bar{u}}{\bar{\zeta}_L}\hat{\zeta}_L\right) dr + \frac{\bar{\zeta}_L}{\bar{u}}\sqrt{\bar{u}^2 - \bar{\zeta}_L^2}\ d\hat{\xi} = 0$$

Let $dM = P\,dr + Q\,d\hat{\xi} = 0$

where $P = i\omega \hat{\xi} - \hat{u} + \frac{\bar{u}}{\bar{\zeta}_L}\hat{\zeta}_L \longrightarrow f(\hat{\xi}, r)$

$Q = \frac{\bar{\zeta}_L}{\bar{u}}\sqrt{\bar{u}^2 - \bar{\zeta}_L^2}$ is constant

$$\Rightarrow \quad \frac{\partial P}{\partial \hat{\xi}} = i\omega \neq 0 = \frac{\partial Q}{\partial r}$$

this differential equation is not a appropriate equation

$\Rightarrow$ Construct a non-zero, derivable function $\mu(r)$, such that satisfies

$$\begin{cases} P\mu\,dr + Q\mu\,d\hat{\xi} = 0 \\ \dfrac{\partial P\mu}{\partial \hat{\xi}} = \dfrac{\partial Q\mu}{\partial r} \end{cases}$$

$$\Rightarrow \quad \frac{\partial P}{\partial \hat{\xi}}\mu + P\frac{\partial \mu}{\partial \hat{\xi}} = \frac{\partial Q}{\partial r}\mu + Q\frac{\partial \mu}{\partial r}$$

$$\Rightarrow \quad i\omega \mu = Q \frac{\partial \mu}{\partial r}$$

$$\Rightarrow \quad \frac{\partial \mu}{\partial r} = \frac{i\omega \mu}{Q}$$

$$\Rightarrow \quad \mu = e^{-\int -\frac{i\omega}{Q} dr} = e^{i\omega r/Q}$$

$$\Rightarrow \quad M = \int Q \mu \, d\hat{\xi}_3 = Q e^{i\omega r/Q} \hat{\xi}_3 + f(r)$$

$$\Rightarrow \quad \frac{\partial M}{\partial r} = i\omega e^{i\omega r/Q} \hat{\xi}_3 + f'(r) = P\mu$$

$$= \left(i\omega \hat{\xi}_3 - \hat{u} + \frac{\bar{u}}{\bar{S}_L} \hat{S}_L \right) e^{i\omega r/Q}$$

$$\Rightarrow \quad f'(r) = \left(-\hat{u} + \frac{\bar{u}}{\bar{S}_L} \hat{S}_L \right) e^{i\omega r/Q}$$

$$\downarrow$$

$$\bar{\xi}_3(r) = (r - R_i) \tan\alpha$$

$$\hat{u} = \varepsilon_u \bar{u} e^{-ik\bar{\xi}_3(r)} = \varepsilon_u \bar{u} e^{-ik(r-R_i)\tan\alpha}$$

$$\hat{S}_L = \varepsilon_{S_L} \bar{S}_L e^{-k_\phi \bar{\xi}_3(r)} = \varepsilon_{S_L} \bar{S}_L e^{-k_\phi (r-R_i)\tan\alpha}$$

$$= \left(-\varepsilon_u \bar{u} e^{-ik(r-R_i)\tan\alpha} + \varepsilon_{S_L} \bar{u} e^{-k_\phi (r-R_i)\tan\alpha} \right) e^{i\omega r/Q}$$

$$= \left(-\varepsilon_u \bar{u} e^{ikR_i\tan\alpha} + \varepsilon_{S_L} \bar{u} e^{-\sigma_\phi r \tan\alpha} e^{k_\phi R_i \tan\alpha} \right)$$

$$\cdot\ e^{-ikr\tan\alpha + i\omega r/Q}$$

$$\Rightarrow f(r) = \frac{-\varepsilon_u \bar{u}\, e^{-ik(r-R_i)\tan\alpha + i\omega r/Q}}{-ik\tan\alpha + i\omega/Q} + \frac{\varepsilon_{SL}\bar{u}\, e^{-k_\phi (r-R_i)\tan\alpha + i\omega r/Q}}{-\sigma_\phi \tan\alpha - ik\tan\alpha + i\omega/Q} + C$$

And $\quad Q e^{i\omega r/Q}\hat{\zeta} + f(r) = 0$

And B.C.: $\hat{\zeta}(R_i) = 0$ (anchored point)

$$\Rightarrow f(R_i) = 0$$

$$\Rightarrow \frac{-\varepsilon_u \bar{u}\, e^{i\omega R_i/Q}}{-ik\tan\alpha + i\omega/Q} + \frac{\varepsilon_{SL}\bar{u}\, e^{i\omega R_i/Q}}{-\sigma_\phi\tan\alpha - ik\tan\alpha + i\omega/Q} + C = 0$$

$$\Rightarrow C = \frac{\varepsilon_u \bar{u}\, e^{i\omega R_i/Q}}{-ik\tan\alpha + i\omega/Q} - \frac{\varepsilon_{SL}\bar{u}\, e^{i\omega R_i/Q}}{-\sigma_\phi\tan\alpha - ik\tan\alpha + i\omega/Q}$$

$$\Rightarrow \hat{\zeta} = \frac{f(r)}{Q e^{i\omega r/Q}} = \frac{-\varepsilon_u \bar{u}\, e^{-ik(r-R_i)\tan\alpha}}{-ikQ\tan\alpha + i\omega} + \frac{\varepsilon_{SL}\bar{u}\, e^{-k_\phi (r-R_i)\tan\alpha}}{-\sigma_\phi Q\tan\alpha - ikQ\tan\alpha + i\omega}$$

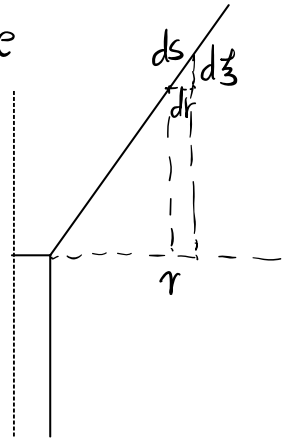
$$+\frac{\varepsilon_u \bar{u}\, e^{-i\omega(r-R_i)/Q}}{-ikQ\tan\alpha + i\omega}$$

$$-\frac{\varepsilon_{SL}\bar{u}\, e^{-i\omega(r-R_i)/Q}}{-\sigma_\phi Q\tan\alpha - ikQ\tan\alpha + i\omega}$$

$$= \frac{\varepsilon_u \bar{u}}{-ikQ\tan\alpha + i\omega}\left(-e^{-ik(r-R_i)\tan\alpha} + e^{-i\omega(r-R_i)/Q}\right)$$

$$+\frac{\varepsilon_{SL}\bar{u}}{-\sigma_\phi Q\tan\alpha - ikQ\tan\alpha + i\omega}\left(e^{-k_\phi(r-R_i)\tan\alpha} - e^{-i\omega(r-R_i)/Q}\right)$$

③ flame surface



$$\Rightarrow \quad ds = \sqrt{(dr)^2 + (dz)^2}$$

$$dA = 2\pi r\, ds = 2\pi r\sqrt{(dr)^2+(dz)^2} = 2\pi r\sqrt{1+\left(\frac{dz}{dr}\right)^2}\, dr$$

$$\Rightarrow \quad \bar{A} = \int d\bar{A} = \int_{R_i}^{R_0} 2\pi r\sqrt{1+\left(\frac{d\bar{z}}{dr}\right)^2}\, dr$$

$$\downarrow \quad \text{steady equation}: \quad 0 = \bar{u} - \bar{S}_L\sqrt{1+\left(\frac{\partial \bar{z}}{\partial r}\right)^2}$$

$$= \int_{R_i}^{R_0} 2\pi r \frac{\bar{u}}{\bar{S_L}}\, dr = \frac{\bar{u}}{\bar{S_L}} \pi \left(R_0^2 - R_i^2\right)$$

$$\Rightarrow \quad \hat{A} = \int_{R_i}^{R_0} 2\pi r \frac{\partial \hat{\xi}}{\partial r} \frac{\partial \bar{\xi}}{\partial r} \frac{1}{\sqrt{1 + \left(\frac{\partial \bar{\xi}}{\partial r}\right)^2}}\, dr$$

$$\downarrow \quad \sqrt{1 + \left(\frac{\partial \bar{\xi}}{\partial r}\right)^2} = \frac{\bar{u}}{\bar{S_L}}$$

$$\frac{\partial \bar{\xi}}{\partial r} = \sqrt{\left(\frac{\bar{u}}{\bar{S_L}}\right)^2 - 1}$$

$$= 2\pi \int_{R_i}^{R_0} r \frac{\partial \hat{\xi}}{\partial r} \sqrt{\left(\frac{\bar{u}}{\bar{S_L}}\right)^2 - 1} \frac{\bar{S_L}}{\bar{u}}\, dr$$

$$= 2\pi \sin\alpha \int_{R_i}^{R_0} r \frac{\partial \hat{\xi}}{\partial r}\, dr$$

$$= 2\pi \sin\alpha \left(r \hat{\xi} \Big|_{R_i}^{R_0} - \int_{R_i}^{R_0} \hat{\xi}\, dr \right)$$

$$= 2\pi \sin\alpha \left(I_1 - I_2 \right)$$

where $I_1 = r \hat{\xi} \Big|_{R_i}^{R_0}$

$$= R_0 \left[\frac{\varepsilon_u \bar{u}}{-ikQ\tan\alpha + i\omega} \left(-e^{-ik(R_0 - R_i)\tan\alpha} + e^{-i\omega(R_0 - R_i)/Q} \right) \right.$$

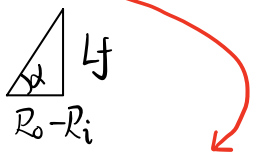
$$\left. + \frac{\varepsilon_L \bar{u}}{-\sigma_\phi Q \tan\alpha - ikQ\tan\alpha + i\omega} \left(e^{-k_\phi (R_0 - R_i)\tan\alpha} \right. \right.$$

$$- e^{-i\omega(R_o - R_i)/Q} \Big) \Big]$$

And $Q = \frac{\bar{\zeta}_L}{\bar{u}} \sqrt{\bar{u}^2 - \bar{\zeta}_L^2} = \bar{\zeta}_L \sqrt{1 - \left(\frac{\bar{\zeta}_L}{\bar{u}}\right)^2} = \bar{\zeta}_L \sin\alpha$

$\Rightarrow$ 1° $ikQ\tan\alpha = i\frac{\omega}{\bar{u}} \bar{\zeta}_L \sin\alpha \tan\alpha = i\omega \sin^2\alpha$

2° we know that $\zeta_t = \frac{2\pi f L_f}{\bar{u}} = \frac{\omega L_f}{\bar{u}} = k(R_o - R_i)\tan\alpha$



$\Rightarrow ik(R_o - R_i)\tan\alpha = i\omega^*$

3° $\sigma_\phi Q \tan\alpha = \sigma_\phi \bar{\zeta}_L \sin\alpha \tan\alpha = \sigma_\phi \bar{u} \frac{\bar{\zeta}_L}{\bar{u}} \sin\alpha \tan\alpha$

$= \sigma_\phi \bar{u} \sin^2\alpha$

4° $k_\phi (R_o - R_i)\tan\alpha = \sigma_\phi (R_o - R_i)\tan\alpha + ik(R_o - R_i)\tan\alpha$

$= \sigma_\phi (R_o - R_i)\tan\alpha + i\omega^*$

5° $i\omega(R_o - R_i)/Q = \frac{i\omega(R_o - R_i)}{\zeta_L \sin\alpha} = \frac{i\omega(R_o - R_i)}{\bar{u} \sin\alpha\cos\alpha}$

$= ik(R_o - R_i)\tan\alpha \frac{1}{\sin^2\alpha}$

$= i\omega^* / \sin^2\alpha$

$$\Rightarrow \quad I_1 = R_0 \left[\frac{\varepsilon_u \bar{u}}{-i\omega \sin^2\alpha + i\omega} \left(-e^{-i\omega^*} + e^{-i\omega^*/\sin^2\alpha} \right) \right.$$

$$\left. + \frac{\varepsilon_{SL} \bar{u}}{-\sigma_\phi \bar{u} \sin^2\alpha - i\omega \sin^2\alpha + i\omega} \left(e^{-\sigma_\phi (R_0 - R_i)\tan\alpha} e^{-i\omega^*} - e^{-i\omega^*/\sin^2\alpha} \right) \right]$$

$$= \frac{\varepsilon_u \bar{u} R_0}{i\omega} \frac{1}{1-\sin^2\alpha} \left(-e^{-i\omega^*} + e^{-i\omega^*/\sin^2\alpha} \right)$$

$$+ \frac{\varepsilon_{SL} \bar{u} R_0}{i\omega} \frac{1}{1 - \sin^2\alpha \left(\frac{\sigma_\phi \bar{u}}{i\omega} + 1 \right)} \left(e^{-\sigma_\phi (R_0 - R_i)\tan\alpha} e^{-i\omega^*} - e^{-i\omega^*/\sin^2\alpha} \right)$$

And $I_2 = \int_{R_i}^{R_0} \hat{\xi}\, dr = \frac{\varepsilon_u \bar{u}}{-ikQ\tan\alpha + i\omega} \left(\frac{e^{-ik(r-R_i)\tan\alpha}}{ik\tan\alpha} \Big|_{R_i}^{R_0} - \frac{e^{-i\omega(r-R_i)/Q}}{i\omega/Q} \Big|_{R_i}^{R_0} \right)$

$$+ \frac{\varepsilon_{SL} \bar{u}}{-\sigma_\phi Q\tan\alpha - ikQ\tan\alpha + i\omega} \left(-\frac{e^{-k_\phi (r-R_i)\tan\alpha}}{k_\phi \tan\alpha} \Big|_{R_i}^{R_0} + \frac{e^{-i\omega(r-R_i)/Q}}{i\omega/Q} \Big|_{R_i}^{R_0} \right)$$

$$= \frac{\varepsilon_u \bar{u}}{-ikQ\tan\alpha + i\omega} \left(\frac{e^{-ik(R_0-R_i)\tan\alpha}}{ik\tan\alpha} - \frac{e^{-i\omega(R_0-R_i)/Q}}{i\omega/Q} - \frac{1}{ik\tan\alpha} + \frac{1}{i\omega/Q} \right)$$

$$+ \frac{\varepsilon_{SL} \bar{u}}{-\sigma_\phi Q\tan\alpha - ikQ\tan\alpha + i\omega} \left(-\frac{e^{-k_\phi (R_0-R_i)\tan\alpha}}{k_\phi \tan\alpha} + \frac{e^{-i\omega(R_0-R_i)/Q}}{i\omega/Q} + \frac{1}{k_\phi \tan\alpha} - \frac{1}{i\omega/Q} \right)$$

$$= \frac{\varepsilon_u \bar{u}}{-i\omega \sin^2\alpha + i\omega} \left(\frac{e^{-i\omega^*}}{ik\tan\alpha} - \frac{e^{-i\omega^*/\sin^2\alpha}}{i\omega/(\bar{S}_L \sin\alpha)} - \frac{1}{ik\tan\alpha} + \frac{1}{i\omega/(\bar{S}_L \sin\alpha)} \right)$$

$$+ \frac{\varepsilon_{SL} \bar{u}}{-\sigma_\phi \bar{u} \sin^2\alpha - i\omega \sin^2\alpha + i\omega} \left(- \frac{e^{-\sigma_\phi (R_0 - R_i)\tan\alpha} e^{-i\omega^*}}{\sigma_\phi \tan\alpha + ik\tan\alpha} + \frac{e^{-i\omega^*/\sin^2\alpha}}{i\omega/(\bar{S}_L \sin\alpha)} \right.$$

$$\left. + \frac{1}{\sigma_\phi \tan\alpha + ik\tan\alpha} - \frac{1}{i\omega/(\bar{S}_L \sin\alpha)} \right)$$

$$= \frac{\varepsilon_u \bar{u}}{i\omega} \frac{1}{1 - \sin^2\alpha} \left(\frac{e^{-i\omega^*}}{ik\tan\alpha} - \frac{e^{-i\omega^*/\sin^2\alpha}}{ik\tan\alpha/\sin^2\alpha} - \frac{1}{ik\tan\alpha} + \frac{1}{ik\tan\alpha/\sin^2\alpha} \right)$$

$$+ \frac{\varepsilon_{SL} \bar{u}}{i\omega} \frac{1}{1 - \sin^2\alpha \left(\frac{\sigma_\phi \bar{u}}{i\omega} + 1 \right)} \left(- \frac{e^{-\sigma_\phi (R_0 - R_i)\tan\alpha} e^{-i\omega^*}}{\sigma_\phi \tan\alpha + ik\tan\alpha} + \frac{e^{-i\omega^*/\sin^2\alpha}}{ik\tan\alpha/\sin^2\alpha} \right.$$

$$\left. + \frac{1}{\sigma_\phi \tan\alpha + ik\tan\alpha} - \frac{1}{ik\tan\alpha/\sin^2\alpha} \right)$$

$$\Rightarrow I_1 - I_2$$

$$= \frac{\varepsilon_u \bar{u}}{i\omega} \frac{1}{1 - \sin^2\alpha} \left[\left(-R_0 - \frac{1}{ik\tan\alpha} \right) e^{-i\omega^*} + \left(R_0 + \frac{1}{ik\tan\alpha/\sin^2\alpha} \right) e^{-i\omega^*/\sin^2\alpha} \right.$$

$$\left. + \frac{1}{ik\tan\alpha} - \frac{1}{ik\tan\alpha/\sin^2\alpha} \right]$$

$$+ \frac{\varepsilon_{SL} \bar{u}}{i\omega} \frac{1}{1 - \sin^2\alpha \left(\frac{\sigma_\phi \bar{u}}{i\omega} + 1 \right)} \left[\left(R_0 + \frac{1}{\sigma_\phi \tan\alpha + ik\tan\alpha} \right) e^{-\sigma_\phi (R_0 - R_i)\tan\alpha} e^{-i\omega^*} \right.$$

$$+\left(-R_0-\frac{1}{ik\tan\alpha/\sin^2\alpha}\right)e^{-i\omega^*/\sin^2\alpha}$$

$$-\frac{1}{\sigma_\phi\tan\alpha+ik\tan\alpha}+\frac{1}{ik\tan\alpha/\sin^2\alpha}\Bigg]$$

$$=\frac{\varepsilon_U}{(i\omega^*)^2}\frac{(R_0-R_i)^2\tan\alpha}{1-\sin^2\alpha}\Bigg[(-ikR_0\tan\alpha-1)e^{-i\omega^*}+(ikR_0\tan\alpha+\sin^2\alpha)e^{-i\omega^*/\sin^2\alpha}$$

$$+1-\sin^2\alpha\Bigg]$$

$$+\frac{\varepsilon_L}{i\omega^*}\frac{(R_0-R_i)\tan\alpha}{1-\sin^2\alpha\left(\frac{\sigma_\phi\bar{u}}{i\omega}+1\right)}\Bigg[\left(R_0+\frac{1}{\sigma_\phi\tan\alpha+ik\tan\alpha}\right)e^{-\sigma_\phi(R_0-R_i)\tan\alpha}e^{-i\omega^*}$$

$$+\left(-R_0-\frac{1}{ik\tan\alpha/\sin^2\alpha}\right)e^{-i\omega^*/\sin^2\alpha}$$

$$-\frac{1}{\sigma_\phi\tan\alpha+ik\tan\alpha}+\frac{1}{ik\tan\alpha/\sin^2\alpha}\Bigg]$$

$$=\frac{\varepsilon_U}{(i\omega^*)^2}\frac{(R_0-R_i)^2\tan\alpha}{1-\sin^2\alpha}\Bigg[ikR_0\tan\alpha\left(-e^{-i\omega^*}+e^{-i\omega^*/\sin^2\alpha}\right)$$

$$+1-e^{-i\omega^*}+\frac{e^{-i\omega^*/\sin^2\alpha}-1}{1/\sin^2\alpha}\Bigg]$$

$$+\frac{\varepsilon_{sl}}{i\omega^*}\frac{(R_o-R_i)\tan\alpha}{1-\sin^2\alpha\left(\frac{\sigma_\phi\bar{u}}{i\omega}+1\right)}\left[\left(R_o+\frac{1}{\sigma_\phi\tan\alpha+ik\tan\alpha}\right)e^{-\sigma_\phi(R_o-R_i)\tan\alpha}e^{-i\omega^*}\right.$$

$$+\left(-R_o-\frac{1}{ik\tan\alpha/\sin^2\alpha}\right)e^{-i\omega^*/\sin^2\alpha}$$

$$\left.-\frac{1}{\sigma_\phi\tan\alpha+ik\tan\alpha}+\frac{1}{ik\tan\alpha/\sin^2\alpha}\right]$$

$$\Rightarrow \frac{\hat{A}}{\bar{A}}=\frac{2\pi\sin\alpha\,(I_1-I_2)}{\frac{\bar{u}}{\bar{s}_i}\pi\,(R_o^2-R_i^2)}=\frac{2\sin\alpha\cos\alpha\,(I_1-I_2)}{(R_o+R_i)(R_o-R_i)}$$

$$=\frac{\varepsilon_u}{(i\omega^*)^2}\frac{2\sin^2\alpha}{1-\sin^2\alpha}\frac{(R_o-R_i)}{(R_o+R_i)}\left[ikR_o\tan\alpha\left(-e^{-i\omega^*}+e^{-i\omega^*/\sin^2\alpha}\right)\right.$$

$$\left.+1-e^{-i\omega^*}+\frac{e^{-i\omega^*/\sin^2\alpha}-1}{1/\sin^2\alpha}\right]$$

$$+\frac{\varepsilon_{sl}}{i\omega^*}\frac{2\sin^2\alpha}{1-\sin^2\alpha\left(\frac{\sigma_\phi\bar{u}}{i\omega}+1\right)}\frac{1}{R_o+R_i}\left[\left(R_o+\frac{1}{\sigma_\phi\tan\alpha+ik\tan\alpha}\right)e^{-\sigma_\phi(R_o-R_i)\tan\alpha}e^{-i\omega^*}\right.$$

$$+\left(-R_o-\frac{1}{ik\tan\alpha/\sin^2\alpha}\right)e^{-i\omega^*/\sin^2\alpha}$$

$$\left.-\frac{1}{\sigma_\phi\tan\alpha+ik\tan\alpha}+\frac{1}{ik\tan\alpha/\sin^2\alpha}\right]$$

$$\Rightarrow \frac{\hat{A}}{\bar{A}} = \bar{F}_{A,u}\,\varepsilon_u + \bar{F}_{A,s_L}\,\varepsilon_{s_L}$$

✡ Validation : For $R_o >> R_i$

① $$\bar{F}_{A,u} = \frac{1}{(i\omega^*)^2}\,\frac{2\sin^2\alpha}{1-\sin^2\alpha}\left[i\omega^*\left(-e^{-i\omega^*} + e^{-i\omega^*/\sin^2\alpha}\right) + 1 - e^{-i\omega^*} + \frac{e^{-i\omega^*/\sin^2\alpha} - 1}{1/\sin^2\alpha}\right]$$

$$= \frac{1}{(i\omega^*)^2}\,\frac{2\sin^2\alpha}{1-\sin^2\alpha}\left(1 - e^{-i\omega^*} + \frac{e^{-i\omega^*/\sin^2\alpha} - 1}{1/\sin^2\alpha}\right)$$

$$+ \frac{1}{i\omega^*}\,\frac{2\sin^2\alpha}{1-\sin^2\alpha}\left(-e^{-i\omega^*} + e^{-i\omega^*/\sin^2\alpha}\right)$$

where $\omega^* = k R_o \tan\alpha$

In Schuller's CnF,

$$\Omega = \frac{\omega R}{S_2 \cos\alpha}$$

$$\bar{F}_{A,u} = \frac{2}{\Omega^2}\,\frac{1}{1-\cos^2\alpha}\left(e^{i\Omega} - 1 - \frac{e^{i\Omega\cos^2\alpha} - 1}{\cos^2\alpha}\right)$$

$$+ \frac{2i}{\Omega}\,\frac{1}{1-\cos^2\alpha}\left(e^{i\Omega\cos^2\alpha} - e^{i\Omega}\right)$$

$$\Rightarrow \quad \omega^* = k R_0 \tan\alpha = \frac{\omega}{\bar{u}} R_0 \frac{\sin\alpha}{\cos\alpha} = \frac{\omega R_0 \sin\alpha}{S_L} = \frac{\omega R_0}{S_L \sin\alpha} \sin^2\alpha$$

$$= \Omega \sin^2\alpha$$

$$\Rightarrow \bar{F}_{A,u} = \frac{2}{i^2\Omega^2 \sin^2\alpha} \frac{1}{1-\sin^2\alpha} \left(1 - e^{-i\Omega\sin^2\alpha} + \frac{e^{-i\Omega} - 1}{1/\sin^2\alpha} \right)$$

$$+ \frac{2}{i\Omega} \frac{1}{1-\sin^2\alpha} \left(-e^{-i\Omega\sin^2\alpha} + e^{-i\Omega} \right)$$

$$= \frac{2}{\Omega^2} \frac{1}{1-\sin^2\alpha} \left(\frac{e^{-i\Omega\sin^2\alpha} - 1}{\sin^2\alpha} + 1 - e^{-i\Omega} \right)$$

$$+ \frac{2i}{\Omega} \frac{1}{1-\sin^2\alpha} \left(e^{-i\Omega\sin^2\alpha} - e^{-i\Omega} \right)$$

② $\sigma_\phi = 0$,

$$\bar{F}_{A,S_L} = \frac{1}{i\omega^*} \frac{2\sin^2\alpha}{1-\sin^2\alpha} \frac{1}{R_0 + R_i} \left[\left(R_0 + \frac{1}{ik\tan\alpha} \right) e^{-i\omega^*} \right.$$

$$+ \left(-R_0 - \frac{1}{ik\tan\alpha/\sin^2\alpha} \right) e^{-i\omega^*/\sin^2\alpha}$$

$$\left. - \frac{1}{ik\tan\alpha} + \frac{1}{ik\tan\alpha/\sin^2\alpha} \right]$$

$$= \frac{1}{(i\omega^*)^2} \frac{2\sin^2\alpha}{1-\sin^2\alpha} \frac{R_0 - R_i}{R_0 + R_i} \left[(ikR_0\tan\alpha + 1) e^{-i\omega^*} \right.$$

$$+\left(-ikZ_0\tan\alpha-\sin^2\alpha\right)e^{-i\omega^*/m^2\alpha}$$

$$-1+\sin^2\alpha\Big]$$

$$=\frac{1}{(i\omega^*)^2}\frac{2\sin^2\alpha}{1-\sin^2\alpha}\frac{Z_0-Z_1}{Z_0+Z_1}\Big[ikZ_0\tan\alpha\left(e^{-i\omega^*}-e^{-i\omega^*/m^2\alpha}\right)$$

$$+e^{-i\omega^*}-1-\frac{e^{-i\omega^*/m^2\alpha}-1}{1/m^2\alpha}\Big]$$

$$=-F_{A,u}$$

符合 ✓

Convective - diffusive of ϕ'

$$\frac{\partial \phi'}{\partial t} + \bar{u}\frac{\partial \phi'}{\partial x} - \Gamma_{eff}\frac{\partial^2 \phi'}{\partial x^2} = 0$$

Take the Fourier transform to t : $\hat{\phi}(x,\omega) = \int_{-\infty}^{+\infty} \phi'(x,t) e^{i\omega t} dt$

$$\Rightarrow \quad i\omega\hat{\phi} + \bar{u}\frac{\partial \hat{\phi}}{\partial x} - \Gamma_{eff}\frac{\partial^2 \hat{\phi}}{\partial x^2} = 0$$

$$\Rightarrow \quad \frac{\partial^2 \hat{\phi}}{\partial x^2} - \frac{\bar{u}}{\Gamma_{eff}}\frac{\partial \hat{\phi}}{\partial x} - \frac{i\omega}{\Gamma_{eff}} = 0$$

$$\Rightarrow \quad \lambda^2 - \frac{\bar{u}}{\Gamma_{eff}}\lambda - \frac{i\omega}{\Gamma_{eff}} = 0$$

$$\Rightarrow \quad \lambda = \frac{\frac{\bar{u}}{\Gamma_{eff}} \pm \sqrt{\frac{\bar{u}^2}{\Gamma_{eff}^2} + 4\frac{i\omega}{\Gamma_{eff}}}}{2}$$

$$= \frac{\bar{u}}{2\Gamma_{eff}}\left(1 \pm \sqrt{1 + 4i\frac{\omega\Gamma_{eff}}{\bar{u}^2}}\right)$$

$$\Rightarrow \quad \hat{\phi}(x,\omega) = \hat{\phi}(0,\omega)\; e^{\frac{\bar{u}}{2\Gamma_{eff}}\left(1 - \sqrt{1+4i\frac{\omega\Gamma_{eff}}{\bar{u}^2}}\right)x}$$

And $x = \bar{\xi}$, $\bar{\xi} = (r - R_i)\tan\alpha$

$$\Rightarrow \quad \hat{\phi}(r,\omega) = \hat{\phi}_0\; e^{\frac{\bar{u}}{2\Gamma_{eff}}\left(1 - \sqrt{1+4i\frac{\omega\Gamma_{eff}}{\bar{u}^2}}\right)\bar{\xi}}$$

where $\sqrt{1+4i\frac{\omega\Gamma_{eff}}{\bar{u}^2}} = \sqrt{1+ia} \xrightarrow{let} A+Bi$

$$\Rightarrow \quad 1+ia = A^2+2ABi-B^2$$

$$\Rightarrow \quad 1+ia = A^2-B^2+2ABi$$

$$\Rightarrow \quad \begin{cases} A^2-B^2=1 \\ 2AB=a \end{cases} \Rightarrow \begin{cases} A^2=\frac{\sqrt{1+a^2}+1}{2} \\ B^2=\frac{\sqrt{1+a^2}-1}{2} \end{cases}$$

A, B 同号 $\Rightarrow \begin{cases} A=\sqrt{\frac{\sqrt{1+a^2}+1}{2}} \\ B=\sqrt{\frac{\sqrt{1+a^2}-1}{2}} \end{cases}$

$$\Rightarrow \quad \sqrt{1+ai} = \sqrt{\frac{\sqrt{1+a^2}+1}{2}} + i\sqrt{\frac{\sqrt{1+a^2}-1}{2}}$$

$$\frac{\bar{u}}{2\Gamma_{eff}}\left(1-\sqrt{1+4i\frac{\omega\Gamma_{eff}}{\bar{u}^2}}\right) = \frac{\bar{u}}{2\Gamma_{eff}}\left(1-\sqrt{\frac{\sqrt{1+a^2}+1}{2}}-i\sqrt{\frac{\sqrt{1+a^2}-1}{2}}\right)$$

$$= -(\sigma_\phi + ik_\phi)$$

$$\Rightarrow \quad \sigma_\phi = \frac{\bar{u}}{2\Gamma_{eff}}\left(\sqrt{\frac{\sqrt{1+a^2}+1}{2}}-1\right)$$

$$k_\phi = \frac{\bar{u}}{2\Gamma_{eff}}\sqrt{\frac{\sqrt{1+a^2}-1}{2}}$$

where $a = \frac{4\omega\Gamma_{eff}}{\bar{u}^2}$

When w is small, a is small

$$\Rightarrow \sqrt{1+a^2}-1 \approx \frac{1}{2}a^2$$

$$\Rightarrow k_\phi \approx \frac{\bar{u}}{2\Gamma_{eff}}\sqrt{\frac{a^2}{4}} = \frac{\bar{u}\,a}{4\Gamma_{eff}}$$

$$= \frac{\bar{u}}{4\Gamma_{eff}}\,\frac{4w\Gamma_{eff}}{\bar{u}^2} = \frac{w}{\bar{u}} = k$$

$$\Rightarrow \sigma_\phi = \frac{\bar{u}}{2\Gamma_{eff}}\left(\sqrt{\frac{\sqrt{1+a^2}+1}{2}}-1\right), \text{ where } a = \frac{4w\Gamma_{eff}}{\bar{u}^2}$$

$$k_\phi \approx \frac{w}{\bar{u}}$$

Example: EQR modulation ($u' = 0$)

$$\Rightarrow \quad \frac{A'}{\bar{A}} = F_{A,S_L}\, \varepsilon_{S_L}$$

where $\varepsilon_{S_L} = \left| \frac{\hat{S}_L}{\bar{S}_L} \right| = m \left| \frac{\hat{\phi}}{\bar{\phi}} \right|$

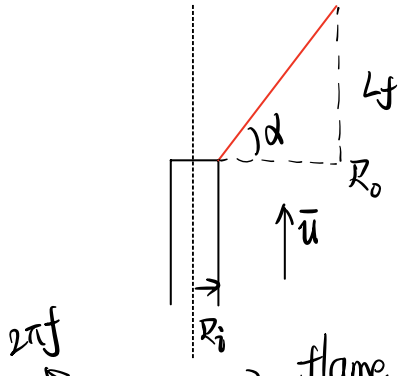
$$F_{A,S_L} =$$

$$\frac{1}{i\omega^*} \frac{2\sin^2\alpha}{1 - \sin^2\alpha\left(\frac{\sigma_\phi \bar{u}}{i\omega} + 1\right)} \frac{1}{R_o + R_i} \left[\left(R_o + \frac{1}{\sigma_\phi \tan\alpha + ik\tan\alpha} \right) e^{-\sigma_\phi (R_o - R_i)\tan\alpha} e^{-i\omega^*} \right.$$

$$+ \left(-R_o - \frac{1}{ik\tan\alpha/\sin^2\alpha} \right) e^{-i\omega^*/\sin^2\alpha}$$

$$\left. - \frac{1}{\sigma_\phi \tan\alpha + ik\tan\alpha} + \frac{1}{ik\tan\alpha/\sin^2\alpha} \right]$$

where



$2\pi f$ ↖ ↗ flame length

$$\omega^* = \frac{\omega L_f}{\bar{u}} = k(R_o - R_i)\tan\alpha$$

$$k = \frac{\omega}{\bar{u}}$$

$$\sigma_\phi = \frac{\bar{u}}{2\Gamma_{eff}} \left(\sqrt{\frac{\sqrt{1+a^2}+1}{2}} - 1 \right), \text{ where } a = \frac{4\omega\Gamma_{eff}}{\bar{u}^2}$$

↓ Diffusion coefficient

化简：

$$F_{A,SL} = \frac{1}{i\omega^*} \frac{2\sin^2\alpha}{1-\sin^2\alpha\left(\frac{\sigma_\phi^*}{i\omega^*}+1\right)} \frac{1}{R_0+R_i} \left[\left(R_0 + \frac{R_0-R_i}{\sigma_\phi^*+i\omega^*}\right) e^{-\sigma_\phi^*} e^{-i\omega^*} \right.$$

$$+ \left(-R_0 - \frac{R_0-R_i}{i\omega^*/\sin^2\alpha}\right) e^{-i\omega^*/\sin^2\alpha}$$

$$\left. - \frac{R_0-R_i}{\sigma_\phi^*+i\omega^*} + \frac{R_0-R_i}{i\omega^*/\sin^2\alpha} \right]$$

$$= \frac{1}{i\omega^*} \frac{2\sin^2\alpha}{1-\sin^2\alpha\left(\frac{\sigma_\phi^*}{i\omega^*}+1\right)} \frac{R_0-R_i}{R_0+R_i} \left[\frac{1}{\sigma_\phi^*+i\omega^*} \left(e^{-\sigma_\phi^*} e^{-i\omega^*} - 1 \right) \right.$$

$$+ \frac{R_0}{R_0-R_i} \left(e^{-\sigma_\phi^*} e^{-i\omega^*} - e^{-i\omega^*/\sin^2\alpha} \right)$$

$$\left. + \frac{1}{i\omega^*/\sin^2\alpha} \left(1 - e^{-i\omega^*/\sin^2\alpha} \right) \right]$$

If $R_0 \gg R_i$

$$\Rightarrow \bar{F}_{A,SL} = \frac{1}{i\omega^*} \frac{2\sin^2\alpha}{1-\sin^2\alpha\left(\frac{\sigma_\phi^*}{i\omega^*}+1\right)} \left[\frac{1}{\sigma_\phi^* + i\omega^*} \left(e^{-\sigma_\phi^*} e^{-i\omega^*} - 1 \right) \right.$$

$$+ \left(e^{-\sigma_\phi^*} e^{-i\omega^*} - e^{-i\omega^*/\sin^2\alpha} \right)$$

$$\left. + \frac{1}{i\omega^*/\sin^2\alpha} \left(1 - e^{-i\omega^*/\sin^2\alpha} \right) \right]$$

where $\sigma_\phi^* = \sigma_\phi (R_0 - R_i) \tan\alpha = \sigma_\phi L_f$

$$\sigma_\phi = \frac{\bar{u}}{2\Gamma_{eff}} \left(\sqrt{\frac{\sqrt{1+a^2}+1}{2}} - 1 \right), \text{ where } a = \frac{4\omega\Gamma_{eff}}{\bar{u}^2}$$

$$\frac{\int \hat{S}_L \, d\bar{A}}{\bar{S}_L \bar{A}} = \frac{\int \frac{d(S_L/\bar{S}_L)}{d(\phi/\bar{\phi})} \frac{\hat{\phi}}{\bar{\phi}} \, d\bar{A}}{\bar{A}}$$

$$= \frac{m}{\bar{A}} \int \frac{\hat{\phi}}{\bar{\phi}} \, d\bar{A}$$

where $m = \frac{\bar{\phi}}{\bar{S}_L} \frac{dS_L}{d\phi}$

And $\hat{\phi}(r) = \varepsilon_\phi \bar{\phi} \, e^{-k_\phi \bar{\xi}(r)}$

$$\Rightarrow \int \frac{\hat{\phi}}{\bar{\phi}} \, d\bar{A} = \int \varepsilon_\phi \, e^{-k_\phi \bar{\xi}(r)} \, d\bar{A}$$

$$= \varepsilon_\phi \int e^{-k_\phi \bar{\xi}(r)} \, d\bar{A}$$

$$= \varepsilon_\phi \int_{R_i}^{R_o} e^{-k_\phi (r - R_i)\tan\alpha} \cdot 2\pi r \sqrt{1 + \left(\frac{d\bar{\xi}}{dr}\right)^2} \, dr$$

$$= \varepsilon_\phi \int_{R_i}^{R_o} e^{-k_\phi (r - R_i)\tan\alpha} \cdot 2\pi r \frac{\bar{u}}{\bar{S}_L} \, dr$$

$$= 2\pi \varepsilon_\phi \frac{\bar{u}}{\bar{S}_L} \int_{R_i}^{R_o} r \, e^{-k_\phi (r - R_i)\tan\alpha} \, dr$$

$$\int u \, dv = uv - \int v \, du$$

$u = r$ (+) $\quad dv = e^{-k_\phi (r - R_i)\tan\alpha}$

$du = 1$ (−) $\quad v = -\frac{1}{k_\phi \tan\alpha} e^{-k_\phi (r - R_i)\tan\alpha}$

$$= 2\pi \varepsilon_\phi \frac{\bar{u}}{\bar{S}_L} \left[-\frac{r}{k_\phi \tan\alpha} e^{-k_\phi (r - R_i)\tan\alpha} \Big|_{R_i}^{R_o} - \frac{1}{(k_\phi \tan\alpha)^2} e^{-k_\phi (r - R_i)\tan\alpha} \Big|_{R_i}^{R_o} \right]$$

$$= 2\pi \varepsilon_\phi \frac{\bar{u}}{\bar{S}_L} \left[-\frac{R_o e^{-k_\phi (R_o - R_i)\tan\alpha}}{k_\phi \tan\alpha} - \frac{1}{(k_\phi \tan\alpha)^2} e^{-k_\phi (R_o - R_i)\tan\alpha} \right.$$

$$+ \frac{R_i}{k_\phi \tan\alpha} + \frac{1}{(k_\phi \tan\alpha)^2} \Bigg]$$

$$= \frac{2\pi \varepsilon_\phi}{\cos\alpha} \frac{1}{k_\phi \tan\alpha} \left(R_i - R_o e^{-k_\phi (R_o - R_i)\tan\alpha} \right)$$

$$+ \frac{2\pi \varepsilon_\phi}{\cos\alpha} \frac{1}{(k_\phi \tan\alpha)^2} \left(1 - e^{-k_\phi (R_o - R_i)\tan\alpha} \right)$$

$$= \frac{2\pi \varepsilon_\phi}{\cos\alpha} \frac{1}{(\sigma_\phi + ik)\tan\alpha} \left(R_i - R_o e^{-\sigma_\phi (R_o - R_i)\tan\alpha} e^{-ik(R_o - R_i)\tan\alpha} \right)$$

$$+ \frac{2\pi \varepsilon_\phi}{\cos\alpha} \frac{1}{(\sigma_\phi + ik)^2 \tan^2\alpha} \left(1 - e^{-\sigma_\phi (R_o - R_i)\tan\alpha} e^{-ik(R_o - R_i)\tan\alpha} \right)$$

$\Rightarrow$

$$\int \frac{\hat{\phi}}{\bar{\phi}} d\bar{A} = \boxed{\begin{aligned} &\frac{2\pi \varepsilon_\phi}{\cos\alpha} \frac{(R_o - R_i)}{\sigma_\phi^* + i\omega^*} \left(R_i - R_o e^{-\sigma_\phi^*} e^{-i\omega^*} \right) \\ &+ \frac{2\pi \varepsilon_\phi}{\cos\alpha} \frac{(R_o - R_i)^2}{(\sigma_\phi^* + i\omega^*)^2} \left(1 - e^{-\sigma_\phi^*} e^{-i\omega^*} \right) \end{aligned}}$$

where $\sigma_\phi^* = \sigma_\phi (R_o - R_i)\tan\alpha$

$\omega^* = k (R_o - R_i)\tan\alpha$

$$\Rightarrow \frac{\int \hat{S}_L \, d\bar{A}}{\bar{S}_L \bar{A}} = \frac{m}{\bar{A}} \int \frac{\hat{\phi}}{\bar{\phi}} d\bar{A}$$

$$= \frac{m}{\frac{\bar{u}}{\bar{S}_L} \pi (R_o^2 - R_i^2)} \int \frac{\hat{\phi}}{\bar{\phi}} d\bar{A}$$

$$= \frac{2m\varepsilon_\phi}{R_0+R_i}\frac{1}{\sigma_\phi^* + i\omega^*}\left(R_i - R_0 e^{-\sigma_\phi^*} e^{-i\omega^*}\right)$$

$$+ \frac{2m\varepsilon_\phi}{R_0+R_i}\frac{R_0-R_i}{(\sigma_\phi^* + i\omega^*)^2}\left(1 - e^{-\sigma_\phi^*} e^{-i\omega^*}\right)$$

where $\sigma_\phi^* = \sigma_\phi (R_0 - R_i)\tan\alpha$

$\omega^* = k(R_0 - R_i)\tan\alpha$

Validation: If $R_0 \gg R_i$, $\sigma_\phi = 0$

$$\Rightarrow F_{S_L} = \frac{\int \hat{S}_L \, d\bar{A}}{\bar{S}_L \bar{A}} \Big/ \varepsilon_\phi = -\frac{2m}{i\omega^*} e^{-i\omega^*} + \frac{2m}{(i\omega^*)^2}(1 - e^{-i\omega^*})$$

$$= \frac{2mi}{\omega^*} e^{-i\omega^*} - \frac{2m}{(\omega^*)^2}(1 - e^{-i\omega^*})$$